# Redlining at the National Level Analysis

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## 0.1 Introduction to Redlining Specific Data

In this part of the analysis, we wish to look into how the Home Owners Loan Corporation (HOLC) act of redlining created a deep embeeded system of racial as well as economic segregation that was ultimately resistant to the legislation meant to rectify the impacts of redlining? Here we this section of the project aims to answer this question by analyzing the long-term effects of redlining by creating three distinct but connected time series featuring data on racial composition, homeownership rates, and home & rent value across HOLC neighborhood grades. Our results showcase how the prejudicial redlining policies created long-lasting barriers that hindered the economic prosperity of African American families for generations.

The dataset covers the periods from 1910 to 2010 note that the data is captured every 10 years and it reflects several major moments in the history of America's housing policy. It includes the 1930s when HOLC was created and redlining began to shape mortgage lending by labeling neighborhoods according to their perceived risk hwoever it often restricting access to financing for minority in pariicular African American communities. The data also spans the period of the Great Migration which was a time when many African Americans moved north which therby increasing housing demand in urban areas and as well as the Civil Rights act which act essentially brought an end to legally enforced racial segregation. The time series extends more than 40 years beyond the 1968 Fair Housing Act, which formally outlawed housing discrimination which is the key act for which our upcoming difference-in-dfferences analysis is based on and this and thus it ended redlining as a policy. Even so, the long time horizon and historical context of the dataset make it well suited for examining how redlining affected African American communities and how limited the impact of later anti-discrimination policies has been in addressing those effects.

The visualizations use historical data from the Federal Reserve Economic Data (FRED) database on the aforementioned three metrics. The database standardizes monetary values to 2010 dollars and matches U.S. Census data with corresponding geocoded HOLC Neighborhoods. The visuals were created using Python and included a color scheme depicting the four grades and their respective perceived financial riskiness. The solid lines indicate the primary group/variable (i.e. African Americans or Home Value), while the dashed lines represent the secondary measurement (Foreign Born or Rent).

When these visual work together they show that the impacts of redlining did not fade over time. In the sense, that the racial composition chart highlights how HOLC grades were closely tied to enforced segregation and the Grade D neighborhoods seeing a dramatic concentration of African American residents. The homeownership graphs then show how this segregation translated into long term barriers to wealth building for African American families. While finally the home value graph once again shows that these effects extend beyond differences in property prices. This together allows for a linking of lower housing values to reduced savings opportunities, disparities in educational quality, and the lasting psychological consequences of sustained disinvestment.

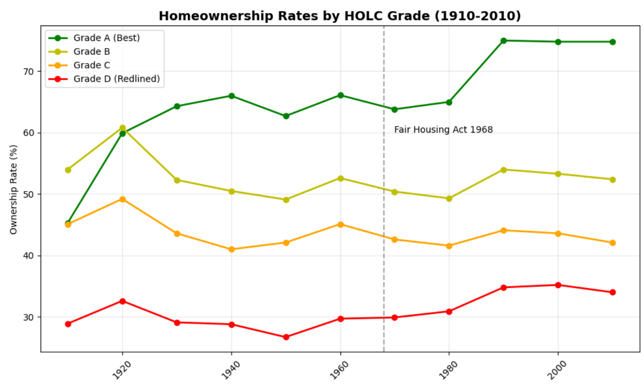
All together these visuals for a nation wide analysis show that by 1968 the resulting inequalities of systemic racism had been so deeply embedded in America's society that ending explicit discrimination through the fair housing act was insufficient. Through this analysis, we hope to call for the need for active intervention as if current policies remain that merley ban redlining the historical patterns will continue to economic prosperity and ultimately opportunity for generations of African Americans and limiting the notion of the American Dream.

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import numpy as np
from pathlib import Path
import statsmodels.api as sm
import warnings
import os
!pip install openpyxl
warnings.filterwarnings('ignore', message='is not UTF -8 encoded')

data_dir = Path("final -group10/data")
```

```
Requirement already satisfied: openpyxl in /srv/conda/envs/notebook/lib/python3.12/site -pac
Requirement already satisfied: et_xmlfile in /srv/conda/envs/notebook/lib/python3.12/site -p
```

```
import matplotlib.pyplot as plt
df_ownership = pd.read_excel("../data/ownership_rate_HOLC.xlsx")
plt.figure(figsize=(10, 6))
plt.plot(df_ownership['observation_date'], df_ownership['A'], 'go -', label='Grade A (Best)'
plt.plot(df_ownership['observation_date'], df_ownership['B'], 'yo -', label='Grade B', linew
plt.plot(df_ownership['observation_date'], df_ownership['C'], 'o -', color='orange', label='
plt.plot(df_ownership['observation_date'], df_ownership['D'], 'ro -', label='Grade D (Redlin
```

```
plt.axvline(pd.to_datetime('1968 -01 -01'), color='gray', linestyle=' - -', alpha=0.7)
plt.text(pd.to_datetime('1970 -01 -01'), 60, 'Fair Housing Act 1968', rotation=0)

plt.title('Homeownership Rates by HOLC Grade (1910 -2010)', fontsize=14, fontweight='bold')
plt.ylabel('Ownership Rate (%)')
plt.legend()
plt.grid(True, alpha=0.3)
plt.xticks(rotation=45)
plt.tight_layout()
plt.savefig('../figures/ownership_simple.png', dpi=300, bbox_inches='tight')
plt.show()
```




The homeownership time series demonstrates the long-lasting barriers induced by redlining that severely limited economic mobility and wealth accumulation.

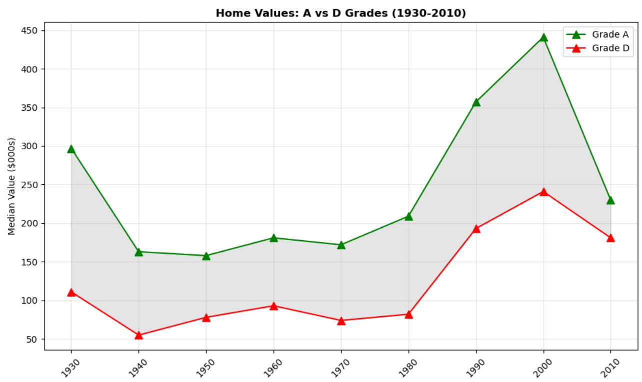
Between the years 1930 to 2010, considerable and consistent gaps emerged between homeownership rates. Residents of Grade A housing were occupied by their owners 75% of the time, while Grade D residents remained consistently suppressed below 35%. This clear contrast represents a systemic rejection of the fundamental wealth-accumulating opportunity for 20th-century Americans.

This alarming 33.4 percentage point between Grade A and D neighborhoods was indicative of the long-lasting nature of redlining, as while White Americans living in Grade A housing were able to accumulate intergenerational wealth through home equity, African American families were trapped in a positive feedback loop of rent cycling.

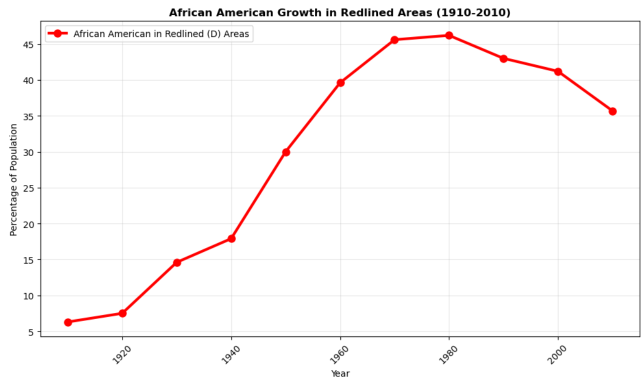
This polarity continued to be reinforced by the contrasting school quality, creating cycles of diverging municipal investment, leaving affluent families unwilling to move in and reside in Grade D and C neighborhoods.

```
df_median = pd.read_excel("../data/median_value_HOLC.xlsx")

plt.figure(figsize=(10, 6))
plt.plot(df_median['observation_date'], df_median['A'], 'g^ -', label='Grade A', markersize=
plt.plot(df_median['observation_date'], df_median['D'], 'r^ -', label='Grade D', markersize=
```

```
plt.fill_between(df_median['observation_date'], df_median['A'], df_median['D'], alpha=0.2, c

plt.title('Home Values: A vs D Grades (1930 -2010)', fontweight='bold')
plt.ylabel('Median Value ($000s)')
plt.legend()
plt.grid(True, alpha=0.3)
plt.xticks(rotation=45)
plt.tight_layout()
plt.savefig('../figures/home_values_gap.png', dpi=300)
plt.show()
```

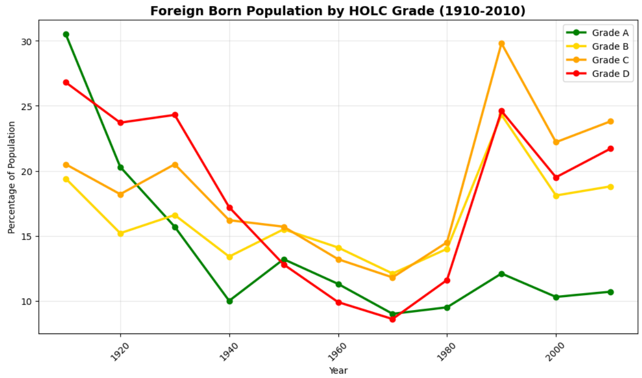



```
# Clean the messy headers skip first row use real column names
df_demo = pd.read_excel("../data/share_of_African_Americans_and_Foreign_Born_HOLC.xlsx",
                        skiprows=1)  # Skip the frequency 10 Year row
df_demo.columns = ['date', 'FB_D', 'FB_A', 'FB_B', 'FB_C', 'AA_A', 'AA_B', 'AA_C', 'AA_D']

plt.figure(figsize=(10, 6))
plt.plot(df_demo['date'], df_demo['AA_D'], 'ro -', linewidth=3, markersize=8,
         label='African American in Redlined (D) Areas')

plt.title('African American Growth in Redlined Areas (1910 -2010)', fontweight='bold')
plt.xlabel('Year')
plt.ylabel('Percentage of Population')
plt.legend()
plt.grid(True, alpha=0.3)
plt.xticks(rotation=45)
plt.tight_layout()
plt.savefig('../figures/aa_redlined_growth.png', dpi=300)
plt.show()
```

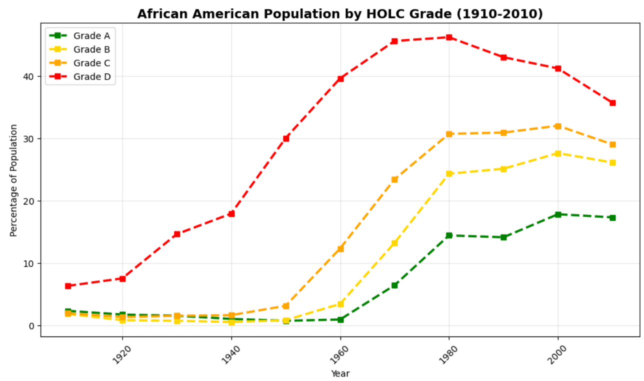



Here we see a distnict increase in the percentage of the population of African Americans in redlined areas, the increase post the 1930s implementation of HOLC redlining is clrease and remains persistent showcasing how African Americans were essentially pushed out from Grade A neighborhoods. The subsequent figures explain the mechanisms behind this increase and compare the concentration of African Americans in redlined vs non-redlined neighborhoods to Foreign Born Americans.

```
plt.figure(figsize=(10, 6))
grade_colors = {'A': 'green', 'B': 'gold', 'C': 'orange', 'D': 'red'}

for grade in ['A', 'B', 'C', 'D']:
    col = f"FB_{grade}"
    plt.plot(df_demo['date'], df_demo[col],
             marker='o', linestyle=' -', color=grade_colors[grade],
             linewidth=2.5, markersize=6, label=f'Grade {grade}')

plt.title('Foreign Born Population by HOLC Grade (1910 -2010)', fontweight='bold', fontsize=
plt.xlabel('Year')
plt.ylabel('Percentage of Population')
plt.legend()
plt.grid(True, alpha=0.3)
plt.xticks(rotation=45)
plt.tight_layout()
plt.savefig('../figures/foreign_born_by_grade.png', dpi=300, bbox_inches='tight')
plt.show()
```




```
plt.figure(figsize=(10, 6))
grade_colors = {'A': 'green', 'B': 'gold', 'C': 'orange', 'D': 'red'}

for grade in ['A', 'B', 'C', 'D']:
    col = f"AA_{grade}"
    plt.plot(
        df_demo['date'],
        df_demo[col],
        marker='s',
        linestyle=' - -',
        color=grade_colors[grade],
        linewidth=2.5,
        markersize=6,
        label=f'Grade {grade}'
    )

plt.title('African American Population by HOLC Grade (1910 -2010)', fontweight='bold', fonts
plt.xlabel('Year')
plt.ylabel('Percentage of Population')
plt.legend()
plt.grid(True, alpha=0.3)
plt.xticks(rotation=45)
plt.tight_layout()
plt.savefig('../figures/african_american_by_grade.png', dpi=300, bbox_inches='tight')
plt.show()
```



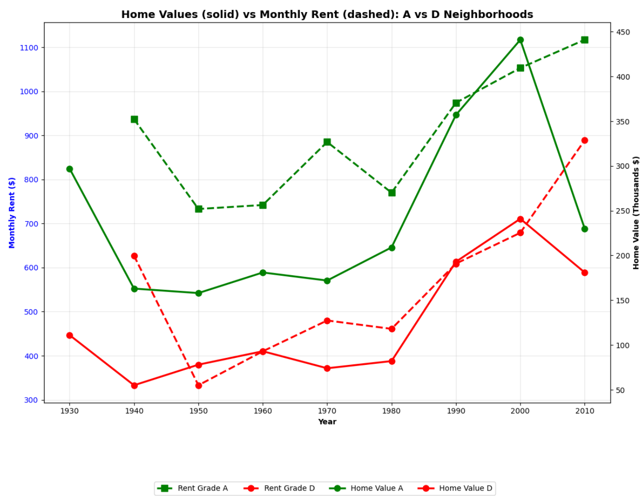

These two figures which show the changes in racial composition over time for African Amercan and foreign-born populations are central to understanding how redlining transformed the racial prejudice into long lasting segregation. In Grade D neighborhoods, the share of African American residents rose sharply from about 6% to 43% between 1910 and 1990, while Grade A neighborhoods experienced a much smaller increase shown from roughly 2% to 14%. This contrast shows how redlining became embedded into racial separation across neighborhoods. The growing gap also reflects the self reinforcing effects of blockbusting which intensified white flight and further restricted African Americans' access to financial services which increased wealth inequality over time.

In contrast the neighborhood differences in the share of foreign born residents who were predomiantly White European were far smaller. In 1970 the range across neighborhoods was only about 3.5 percentage points to nearly 39 percentage points for African Americans. This suggests that European immigrants experienced greater residenial mobility while African Americans remained constrained. Both these time serties highlight how redlining shaped evolving definitions of who was considered white and who was granted access to the benefits associated with whiteness. The figures show that the passage of the Fair Housing Act and other civil rights legislation did little to undo segregation across HOLC graded neighborhoods. The persistence of these patterns underscores how deeply embedded the effects of redlining were and suggests that addressing residential inequality needs more than formal anti discrimination laws it demands active sustained intervention to counter decades of institutionalized discrimination.

```
# Clean rent data (A vs D only)
df_rent = pd.read_excel("../data/rent_and_rental_costs_in_HOLC.xls", skiprows=1)
df_rent.columns = ['date', 'rent_A', 'rent_D']

fig, ax1 = plt.subplots(figsize=(12, 8))

grade_colors = {'A': 'green', 'D': 'red'}

# Rent is left axis with dashed lines matching neighborhood colors
```

```
ax1.plot(df_rent['date'], df_rent['rent_A'], 'g - -s', linewidth=2.5, markersize=8, label='R
ax1.plot(df_rent['date'], df_rent['rent_D'], 'r - -o', linewidth=2.5, markersize=8, label='R
ax1.set_xlabel('Year', fontweight='bold')
ax1.set_ylabel('Monthly Rent ($)', color='blue', fontweight='bold')
ax1.tick_params(axis='y', labelcolor='blue')
ax1.grid(True, alpha=0.3)

# Home values are the right axis solid lines matching neighborhood colors
ax2 = ax1.twinx()
ax2.plot(df_median['observation_date'], df_median['A'], 'g -o', linewidth=2.5, markersize=8,
ax2.plot(df_median['observation_date'], df_median['D'], 'r -o', linewidth=2.5, markersize=8,
ax2.set_ylabel('Home Value (Thousands $)', color='black', fontweight='bold')
ax2.tick_params(axis='y', labelcolor='black')

plt.title('Home Values (solid) vs Monthly Rent (dashed): A vs D Neighborhoods',
          fontweight='bold', fontsize=14)
fig.legend(loc='lower center', bbox_to_anchor=(0.5, -0.15), ncol=4)
plt.xticks(rotation=45)
plt.tight_layout()
plt.savefig('../figures/home_rent_neighborhood_colors.png', dpi=300, bbox_inches='tight')
plt.show()
```

```
/tmp/ipykernel_1877/354147005.py:6: UserWarning: color is redundantly defined by the 'color'
  ax1.plot(df_rent['date'], df_rent['rent_A'], 'g - -s', linewidth=2.5, markersize=8, label=
/tmp/ipykernel_1877/354147005.py:7: UserWarning: color is redundantly defined by the 'color'
  ax1.plot(df_rent['date'], df_rent['rent_D'], 'r - -o', linewidth=2.5, markersize=8, label=
/tmp/ipykernel_1877/354147005.py:15: UserWarning: color is redundantly defined by the 'color
  ax2.plot(df_median['observation_date'], df_median['A'], 'g -o', linewidth=2.5, markersize=
/tmp/ipykernel_1877/354147005.py:16: UserWarning: color is redundantly defined by the 'color
  ax2.plot(df_median['observation_date'], df_median['D'], 'r -o', linewidth=2.5, markersize=
```



The side-by-side time series of home values and rent points to a quiet but powerful system of wealth suppression that worked through several connected channels. Before any anti-discrimination laws were passed, there was already about a $123,000 gap between home values in Grade A and Grade D neighborhoods. That gap alone made it basically impossible for Grade D households to ever catch up, because even if prices grew at the same rate, the difference would only get larger over time. While residents in Grade D areas paid lower rents in dollar terms, they actually spent a much larger share of their income on rent. In many cases, these families were paying amounts comparable to what a mortgage would cost, yet they were denied access to homeownership and the ability to build equity. This higher rent burden lead to less money left over to save or inves which created long term financial strain and made it harder for wealth to be passed down across generations. At the same time, rent payments were effectively flowing to wealthier property owners, often outside Grade D neighborhoods, which only widened existing inequalities. While also the lower home values also meant a weaker property tax base which then leading to reduced school funding and decreasing education quality in redlined areas. When combined with limited savings for down payments, this made upward mobility made it for generations extremely difficult for Grade D residents. Redlining also had a lasting psychological and market effect. Even after the Fair Housing Act, there was little evidence of meaningful catch-up, which later difference-in-differences results will show more clearly. HOLC risk grades ended up acting like a self-fulfilling prophecy: home values reflected these labels rather than any true underlying risk. As a result, Grade D housing continued to follow patterns set decades earlier, instead of being priced based on real fundamentals. Overall, this time series suggests that redlining didn't just create wealth gaps, but helped lock in a racial hierarchy that persisted through what looked like neutral market outcomes.

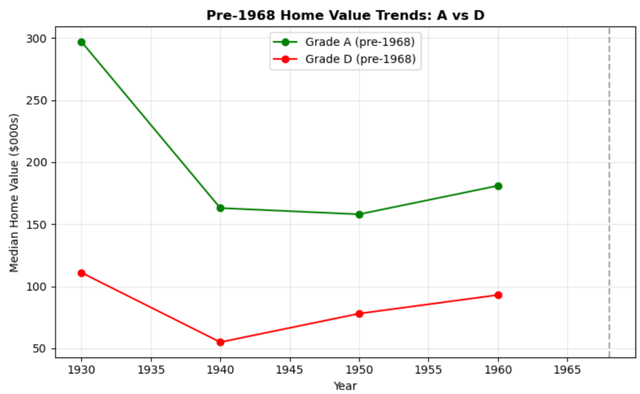
# 1 Difference in Differences Analysis

```
home_did = pd.DataFrame({
    "year": df_median["observation_date"].dt.year,
    "A": df_median["A"],
    "D": df_median["D"]
})

# Keep the common 1930-2010 years
home_did = home_did[(home_did["year"] >= 1930) & (home_did["year"] <= 2010)]

# Put into long (A/D stacked) format
home_long = pd.melt(
    home_did,
    id_vars="year",
    value_vars=["A", "D"],
```

```
    var_name="grade",
    value_name="home_value"
)

home_long["treatment"] = (home_long["grade"] == "D").astype(int)
home_long["post_1968"] = (home_long["year"] >= 1968).astype(int)
home_long["treat_post"] = home_long["treatment"] * home_long["post_1968"]

home_long.head()

home_long.to_csv("home_long.csv", index=False)
```

Here, we prepare the dataset for our difference in differences analysis by first extracting the years from the home value dataset for Grades A and D and then creating the three DiD key variables of treatment (1 for Grade 1 (readline) and 0 for Grade A), post_1968 (true after the Fair housing Act) and treat_post for the interaction term.

```
plt.figure(figsize=(8, 5))

pre = home_did[home_did["year"] < 1968]
post = home_did[home_did["year"] >= 1968]

plt.plot(pre["year"], pre["A"], 'g -o', label='Grade A (pre -1968)')
plt.plot(pre["year"], pre["D"], 'r -o', label='Grade D (pre -1968)')
plt.axvline(1968, color='gray', linestyle=' - -', alpha=0.7)

plt.title('Pre -1968 Home Value Trends: A vs D', fontweight='bold')
plt.xlabel('Year')
plt.ylabel('Median Home Value ($000s)')
plt.legend()
plt.grid(True, alpha=0.3)
plt.tight_layout()
plt.savefig('../figures/home_values_parallel_check.png', dpi=300, bbox_inches='tight')
plt.show()
```




### 1.0.1 Parallel Trends Verification

Here it can be see that before the implementation of the fair housing act, both the Grade A and Grade D neighborhoods have similar trendining values of their respective median home values with a steady decrease in the years 1930-1940 and a slight but steady growth from 1940 to 1960, though there is some varability it can be assume that these trends are reasonably parallel indicating that the differene-in-differences results can be applied.

```
import statsmodels.api as sm

X = home_long[["treatment", "post_1968", "treat_post"]]
X = sm.add_constant(X)
y = home_long["home_value"]

did_model_home = sm.OLS(y, X).fit()
print(did_model_home.summary())
```

```
OLS Regression Results
==============================================================================
Dep. Variable:             home_value   R -squared:                      0.517
Model:                            OLS   Adj. R -squared:                 0.413
Method:                 Least Squares   F -statistic:                    4.990
Date:                Wed, 17 Dec 2025   Prob (F -statistic):            0.0147
Time:                        05:57:33   Log -Likelihood:               -101.89
No. Observations:                  18   AIC:                            211.8
Df Residuals:                      14   BIC:                            215.3
Df Model:                           3
Covariance Type:            nonrobust
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
 - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -
const        199.7500     39.422      5.067      0.000     115.197     284.303
```

```
treatment   -115.5000     55.752     -2.072      0.057    -235.076       4.076
post_1968     82.0500     52.891      1.551      0.143     -31.389     195.489
treat_post   -12.1000     74.799     -0.162      0.874    -172.527     148.327
==============================================================================
Omnibus:                        1.278   Durbin -Watson:                  1.585
Prob(Omnibus):                  0.528   Jarque -Bera (JB):               1.001
Skew:                           0.535   Prob(JB):                        0.606
Kurtosis:                       2.564   Cond. No.                         7.25
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

/srv/conda/envs/notebook/lib/python3.12/site -packages/scipy/stats/_axis_nan_policy.py:430:
  return hypotest_fun_in(*args, **kwds)
```

The difference-in-differene analysis show that the 1968 Fair housing act had no statisically significant impact on the economic dividive between Grade D (redlined) and Grade A (non-redlinied) neighborhoods. The interaction term of treat_post had a coefficient of -12.1 and was insignificant with a p-value of 0.874 indicating that the home values in redline areas to did not increase in value to the values of the Grade A neighborhoods after 1969 beyond from the intial baseline gap of -115.5 and the economy wide post 1968 gap of a +82.1. In context, this means that redlined homes continued to remain over $100,000 behind the Grade A neighborhoods in their median value even after 2 decades of the policy had past and there is still no evidence of the gap narrowing. This shows that despite processive legislation, such as the Fair Housing Act which explicititly banned the HOLC's discriminatory redlining policy, redlinining had already begun to create a self-fullfilling cycle of disinvestment of infastructure leading to worse schools and lowered propperty values stemming from the lower tax revenues. Thus, the statistical insignificance of the difference in differences show that the anti-distriminatory legislation alone did not combat the persistent inequality posed by redlining.